



Gas Leakage Detection System

For

Sensor Lab mini Project

**Third Year (Semester-VI) Bachelors in Information Technology, Autonomous
Program**

By

Student Name: Madhura Jangale

Class: D15A

Roll no:20

Student Name: Omkar Gholap

Class: D15A

Roll no:17

Student Name: Aditya Sampath Kumar

Class: D15A

Roll no:1



_Department of Information Technology

**Vivekanand Education Society's Institute of Technology
An Autonomous Institute Affiliated to University of Mumbai**

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Certificate

This is to certify that **Madhura Jangale (20), Omkar Gholap (17), and Aditya Sampath Kumar (1)** have completed the project report on the topic **Gas Leakage Detection System** satisfactorily in partial fulfilment of the requirements for the award of Mini Project in sensor lab of third Year, (Semester-VI) in Information Technology under the guidance of Mr. Abhishek Chaudhari during the year 2023-2024 as prescribed by An Autonomous Institute Affiliated to University of Mumbai.

Supervisor/ Examiner

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Abstract

The Gas Detection System using MQ5 sensor is a safety-oriented project aimed at detecting harmful gas leaks in the environment. This embedded system utilizes an MQ5 gas sensor to sense flammable gases such as LPG, methane, and butane. An Arduino Uno microcontroller processes the analog readings from the sensor and activates an LED indicator if the detected gas concentration crosses a predefined threshold. This project demonstrates a simple yet effective safety application that can be implemented in homes, labs, or industries to help prevent hazardous situations due to gas leaks.

Chapter 1: Introduction

1.1 Overview

Gas leaks pose serious safety hazards including fire risks, explosions, and health issues. Detecting such leaks early is essential to avoid catastrophic accidents. The Gas Detection System using MQ5 is built to detect such harmful gases and alert users before the concentration becomes dangerous. This system is designed using Arduino Uno and MQ5 gas sensor, and uses a visual alert system (LED) to warn the users.

1.2 Working Principle

The gas detection system operates based on a simple yet effective sensing and response mechanism, broken down into the following stages:

1. Sensing:

- The **MQ-5 gas sensor** continuously monitors the surrounding air for the presence of flammable gases.
- It contains a **sensitive layer made of SnO_2 (tin dioxide)**, which reacts with combustible gases, causing a change in its electrical resistance.

2. Signal Processing:

- The variation in resistance due to gas exposure is converted into a corresponding **analog voltage signal**.
- This signal is proportional to the gas concentration in the air.

3. Comparison and Decision Making:

- The analog signal is fed to the **analog input of the Arduino Uno**.
- The Arduino reads the value and compares it against a **predefined threshold level**, which represents a safe gas concentration limit.

4. Alert Mechanism:

- If the measured value **exceeds the threshold**, it indicates a possible gas leak.
- The Arduino then triggers an **LED to turn ON** as a visual alert to the user.
- This simple output can also be expanded to include **buzzers, GSM alerts, or exhaust fans** in more advanced versions.

Chapter 2: Literature Review

1. **Embedded System for Lethal Gas Leakage Exposure and Forewarning System Using ARM Processor**

Description: This paper proposes an embedded safety indication system to detect and alert about the leakage of hazardous gases in semiconductor manufacturing industries. The system utilizes an ARM processor to monitor gas levels and noise deviations, sending immediate alerts to industry representatives via messages, displaying alerts, cutting off the system's power supply, and ringing alarms to indicate accidents. [5]

2. **R. Singh, "Comparison of MQ Series Sensors for Environmental Monitoring," Electronics Today, 2020.**

Singh's work provides a comparative analysis of various MQ-series gas sensors (such as MQ-2, MQ-3, MQ-6, and MQ-135) in terms of sensitivity, power consumption, and gas specificity. The paper is particularly useful in selecting the appropriate sensor based on application-specific needs.

3. **Gas Leakage Detection Based on IoT**

This paper discusses an Internet of Things (IoT)-based gas leakage detection system. The system is designed to monitor gas levels in real-time and provide alerts in case of leakage. Specific details about the sensors used or the system architecture are not provided in the available information.[6]

Research Gap

- **High Cost and Infrastructure Needs:** Most industrial solutions and IoT-based systems involve complex architecture and high implementation costs, making them less feasible for small-scale or low-income users.
- **Internet Dependence:** Systems relying on real-time IoT alerts are not suitable for environments with limited or no internet access.
- **Limited Focus on Minimalistic Design:** There is a scarcity of affordable, standalone systems focused solely on essential features like local alerts (buzzers, LEDs) and basic gas detection using a single MQ sensor.

Chapter 3: Project Description

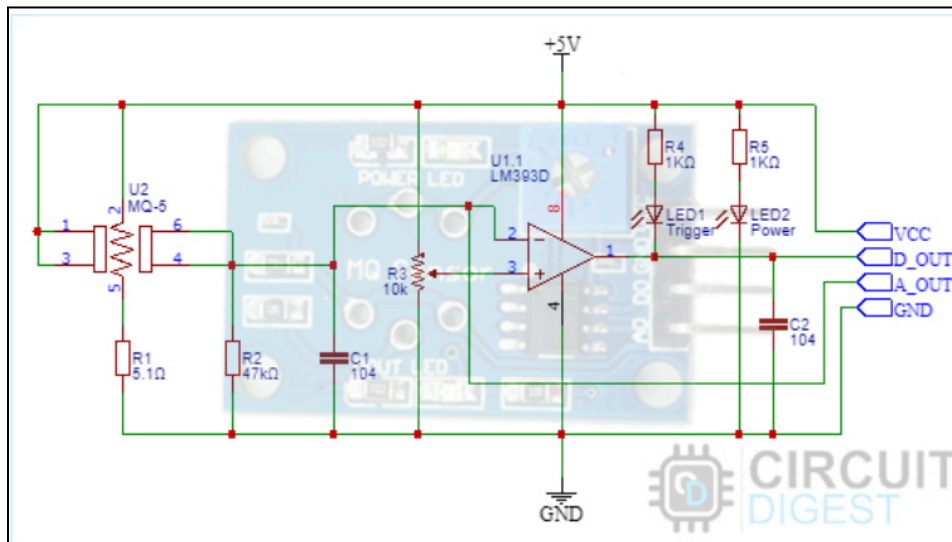
3.1 Problem Definition

There are frequent cases of unnoticed gas leaks which result in dangerous incidents. Human detection is not reliable due to the odourless or mild smell of gases. The need arises for an automated, real-time gas detection mechanism that can operate independently.

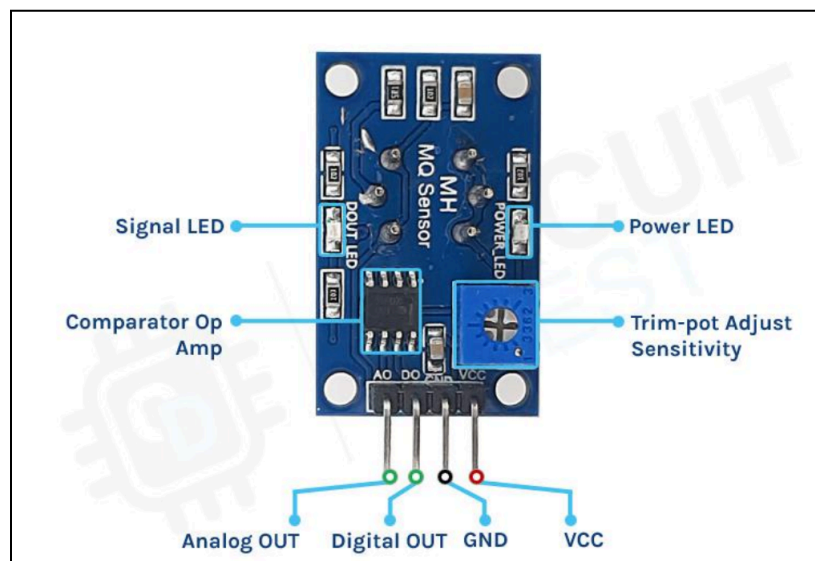
3.2 Steps Involved

1. Component selection
2. Circuit design and connections on breadboard
3. Programming the Arduino
4. Setting the gas detection threshold
5. Testing the system under controlled conditions

3.3 Circuit Diagram



3.4 Component Description



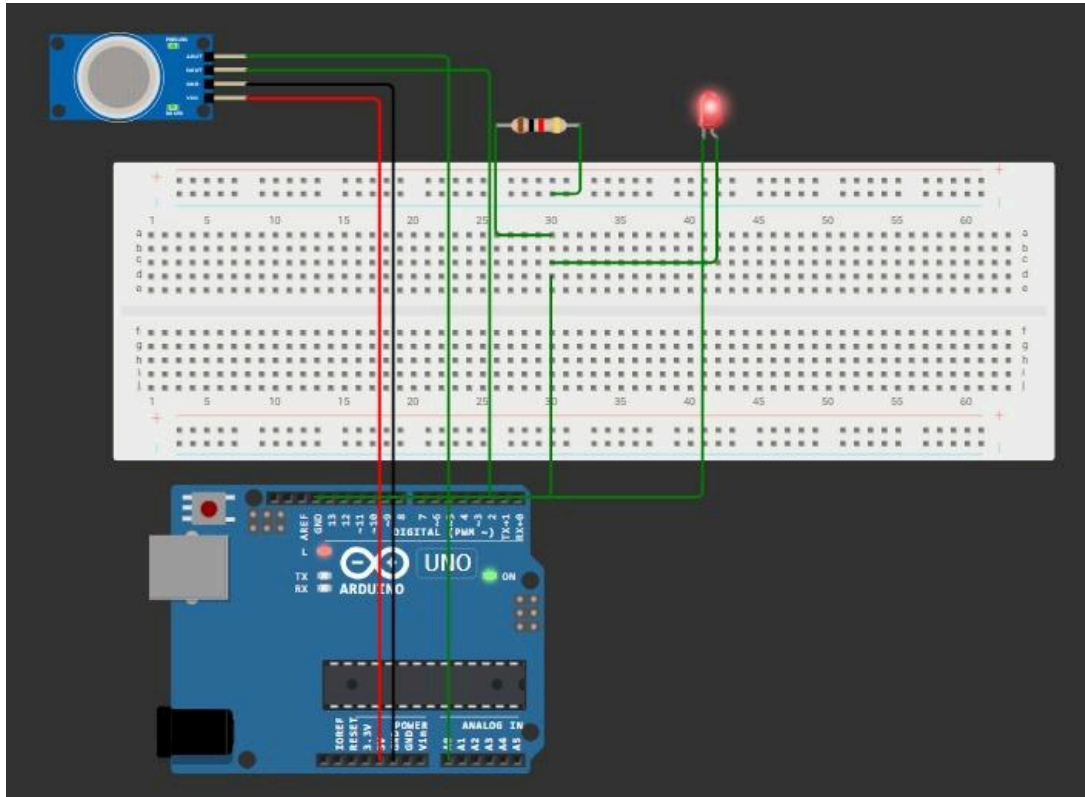
3.4.1 Hardware

- **Arduino Uno**
 - A microcontroller board based on the ATmega328P. It serves as the brain of the system, reading data from the MQ-5 sensor and triggering the alert mechanism based on logic.
- **MQ-5 Gas Sensor**
 - A gas sensor module that detects flammable gases such as LPG, natural gas, and methane. It provides an analog output proportional to the gas concentration in the surrounding air.
- **LED**
 - Used as a **visual alert system** to indicate gas leakage. When the gas level crosses the safe threshold, the LED turns ON.
- **Resistor (220Ω)**
 - Connected in series with the LED to limit the current and prevent damage to the LED.
- **Jumper Wires and Breadboard**
 - Used for making temporary circuit connections during the prototype stage.

3.4.2 Software

- **Arduino IDE**
 - Writing and uploading code
 - Uses analogRead and digitalWrite functions

3.5 Simulation



3.6 Working of the Project

The Gas Leakage Detection System operates in four main stages:

1. Continuous Monitoring:

- The MQ-5 sensor continuously senses the surrounding air for the presence of flammable gases.
- It outputs an analog voltage that increases with the gas concentration.

2. Data Processing:

- This analog signal is read by the Arduino Uno using one of its analog input pins (e.g., A0).
- The Arduino then compares the incoming value to a predefined threshold that indicates a safe gas level.

3. Decision Making:

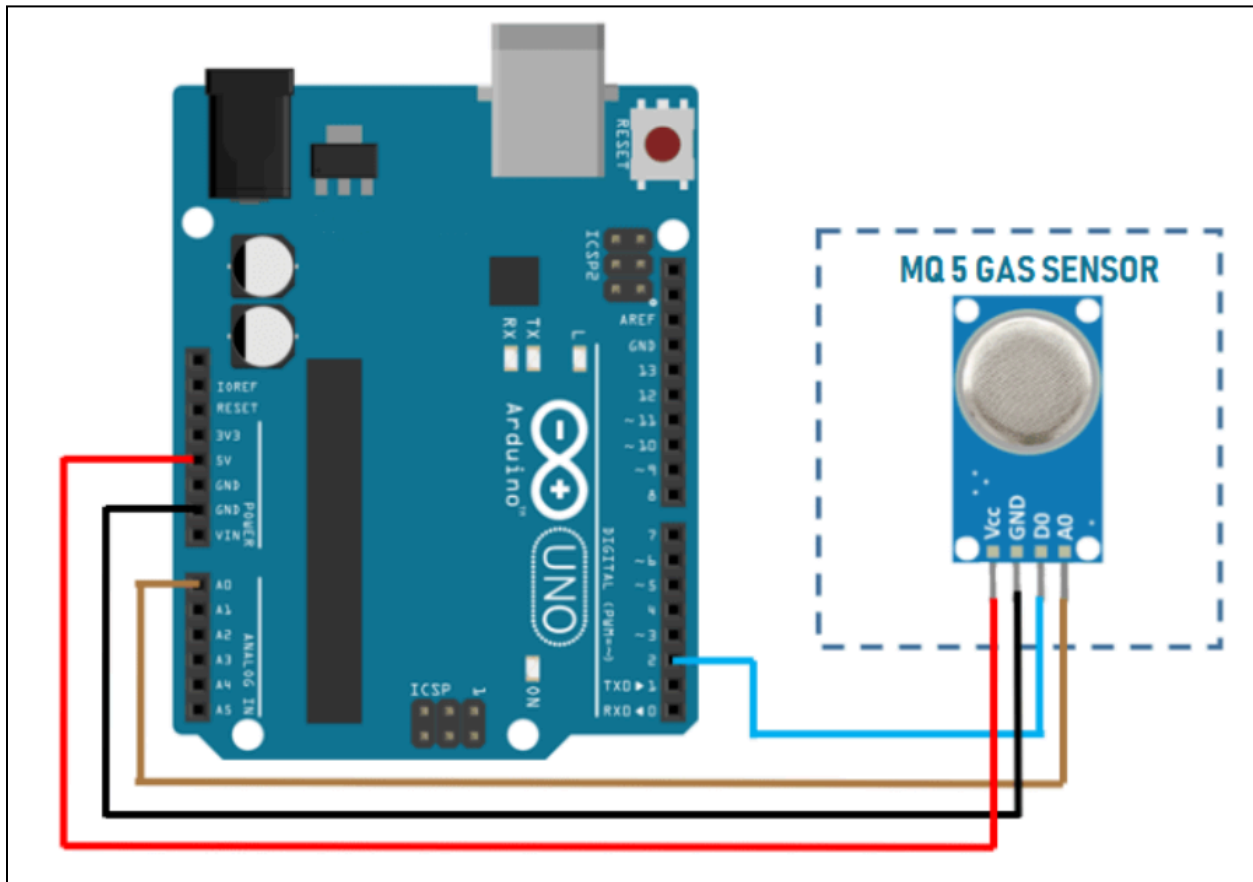
- If the gas concentration is below the threshold, it is considered safe, and the alert system remains OFF.
- If the gas concentration is above the threshold, it indicates a potential leak.

4. Alert Activation:

- When a leak is detected, the Arduino turns ON the LED to alert the user of danger.
- This alert mechanism can be expanded to include buzzers, GSM modules for messaging, or fan activation for ventilation.

This process runs continuously, making the system a real-time gas monitoring and warning system designed for basic safety in kitchens, homes, or small industries.

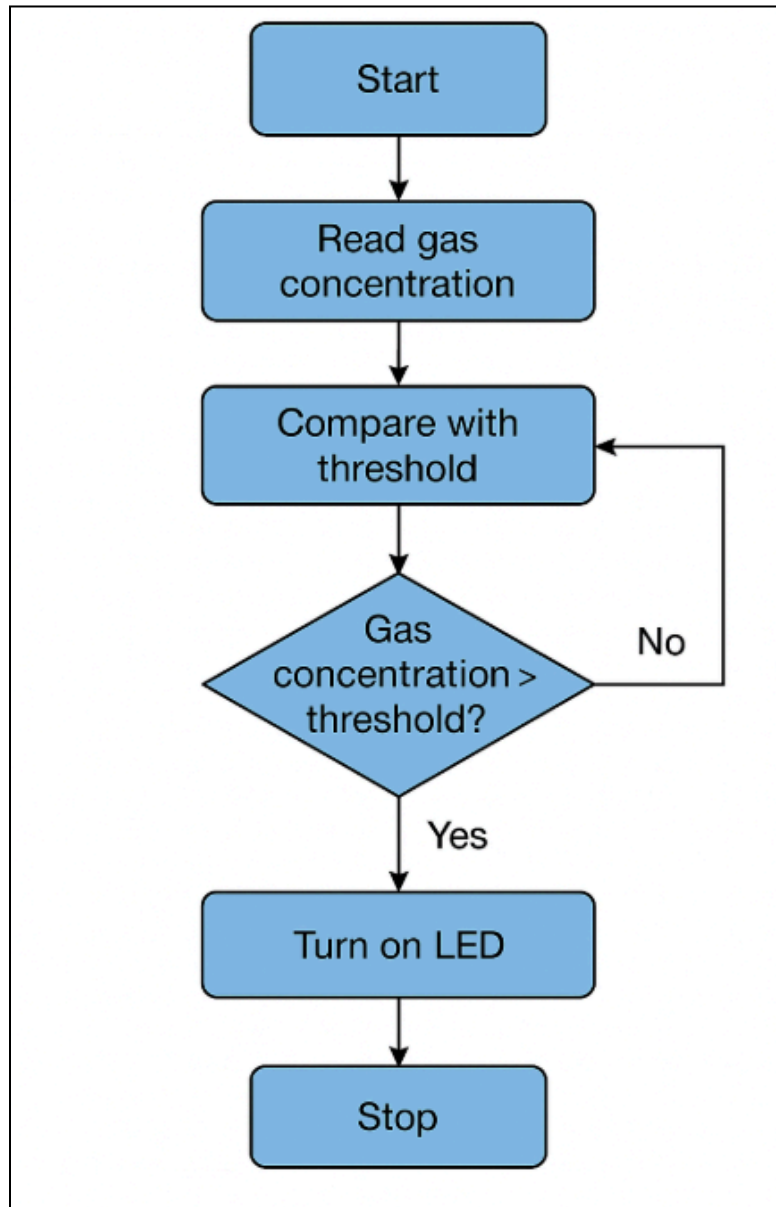
Chapter 4: Implementation



4.1 Hardware Setup

Component	Arduino Pin
MQ5 (VCC)	5V
MQ5 (GND)	GND
MQ5 (A0)	A0
LED (+)	D13
LED (-)	GND

4.2 Code Flow



Algorithm-

Step 1: Start the system.

Step 2: Initialize hardware components:

- Set the **LED pin** as output.
- Set the **analog pin (A0)** for reading gas sensor data.
- Initialize the **serial communication** (optional for monitoring).

Step 3: Define a gas concentration threshold value (based on safe levels).

Step 4: Continuously perform the following loop:

1. **Read analog value** from MQ-5 gas sensor.
2. **Display the sensor reading** on the serial monitor (for debugging).
3. **Compare** the sensor reading with the threshold:
 - If the gas concentration is **less than** the threshold:
→ Turn **OFF** the LED (no gas leak).
 - If the gas concentration is **greater than or equal to** the threshold:
→ Turn **ON** the LED (gas leak detected).
4. **Wait** for a short delay (e.g., 1 second).
5. **Repeat** the loop.

Step 5: End (loop runs continuously).

Code-

```
// scientist BENIEL'S LAB
// GAS DETECTOR

int Input = A0;
int R_LED = 3;

// VAL INTEGER
int val;
int MAX = 300;

void setup() {
  Serial.begin(9600);
  pinMode(Input ,INPUT);
  pinMode(R_LED ,OUTPUT);
}

void loop() {
  val = analogRead(A0);
  Serial.println(val);

  if (val >= MAX) {
    digitalWrite(R_LED , HIGH);

    delay(5000);
    Serial.println("GAS LEAKING");
  }
  else {
    digitalWrite(R_LED ,LOW);

    Serial.println("NORMAL");
    delay(5000);
  }
}
```

Chapter 5: Results and Conclusion

Results & Observations

- **Gas Detection:** The sensor reliably detects butane and LPG when exposed.
- **LED Indication:** LED turns on promptly at predefined threshold.
- **Arduino Performance:** Stable and consistent readings during tests.
- **Calibration:** Different gases yield different analog values; proper tuning was done.

Conclusion

The MQ5 gas detection system is a low-cost, efficient project for detecting harmful gases. It is useful in small environments such as kitchens, labs, or workshops. With minimal components, it provides a quick response visual alert.

Advantages:

- Simple and low-cost
- Visual real-time alert
- Easy to assemble

Future Scope:

- Add a buzzer for sound alert
- Integrate GSM module for SMS alert
- Display real-time gas levels on an LCD

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7. <https://youtu.be/Jwwx00hhztY?feature=shared>